

## HOMEWORK 3

### *Track 2*

Due date: October 23, 2023.

**Reading assignment:** Read chapter 3 of the class notes (on class web-site). Read chapter 4 of the text book. Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

1. Show that if  $A$  has rank  $p$ , then you can find two full column-rank matrices  $X$  and  $Y$  of rank  $p$ , such that  $A = XY^T$ . (Repeat from previous home work.)
2. If  $X$  and  $Y$  are two full-column rank matrices of rank  $p$ , show that the rank of  $XY^T$  is also  $p$ .
3. If  $X$  and  $Y$  are two full column-rank matrices of rank  $p$ , find all matrices  $A$  for which  $R(A) \subseteq R(X)$  and  $R(A^T) \subseteq R(Y)$ .
4. If  $X$  and  $Y$  are two full column-rank matrices of rank  $p$ , find all matrices  $A$  for which  $R(A) = R(X)$  and  $R(A^T) = R(Y)$ .
5. Let  $X$  be a full column-rank matrix. Does there exist an *invertible* matrix  $G$  such that

$$GX = \begin{pmatrix} I \\ 0 \end{pmatrix}?$$

6. Let  $W$ ,  $X$ ,  $Y$  and  $Z$  be full column rank real matrices. What are the conditions (if any) on  $W$ ,  $X$ ,  $Y$  and  $Z$ , such that there exists a real matrix  $A$  such that  $AX = Z$  and  $Y^T A = W^T$ ? Find all such  $A$  when the conditions are satisfied. How many linearly independent equations were there? What was the dimensionality of the left and right nullspaces? Did the rank-nullity theorem hold? The previous exercise can provide a simple way to approach this problem.